

1.2.4 Value

Finally, `value` is what we want the variable to store.

1.3 Primitive data types

Java is a Strongly-typed language. That means, every variable must be declared as a type. In Java, there are 8 primitive types:

- 6 of those refer to numbers
 - 4 for integers types,
 - 2 for floating-point types,
- 1 is the character type `char`, used for characters in Unicode encoding, and
- 1 is a boolean type for true or false values.

1.3.1 `int`

In contrast to C/C++, an `int` will always--no matter which operating system--take 4 bytes of storage space. Because those 4 bytes are set in stone, you can be sure that every JVM that runs your program will be able to store the same size numbers.

`int` is the most commonly used number size.

Range: -2,147,483,648 to 2,147,483,647 (over two billion)

1.3.2 `Short`

In Java, a `short` is defined as 2 bytes, no matter which operating system is used. You would only use this for special situations, such as when speed is really crucial. For VB programmers, a `short` is what you've come to think of as an `int`.

Range: -32,768 to 32,767

1.3.3 `long`

A `long` is defined as 8 bytes, no matter which operating system is used.

Range: -9,223,372,036,854,775,808L to 9,223,372,036,854,775,807L

Please notice the upper-case L suffix is appended to any `long`. This is required.

Hexadecimal numbers have a prefix: `0x`

`0x1CFE`.

1.3.4 `byte`

A `byte` is defined as 1 byte, no matter which operating system is used.